Eco-evolutionary deviations from steady state across the Hawai‘i chronosequence revealed with a maximum entropy perspective

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Introduction

Biodiversity patterns are bound by two seemingly contradictory states: they show striking consistency in form across systems, and yet also must emerge in spite of, or because of, the constant flux and dynamic change of ecology and evolution. The near universality of biodiversity patterns includes the scaling of species richness with area, the ubiquity of rare spaces and scarcity of common species, and the highly skewed distribution of biomass or metabolic resource use across organisms. Debate has persisted for over a century as to whether these consistent patterns form because of ecological and evolutionary drivers that force systems into these configurations, or are largely the outcome of chance acting on large systems of constituent parts. The former gives rise to niche-based theories of coexistence, while that later has produced a number of neutral-like models.

Both deterministic and stochastic perspectives on the generation of universal biodiversity patterns most often make the choice of considering an ecosystem at “equilibrium” or steady state. However, ecosystems are known to be dynamic both through intrinsic eco-evolutionary processes and due to recent anthropogenic perturbation. Whether universal patterns are insensitive to non-steady-state processes or are a result of those processes remains addressed. To better understand the interaction of non-steady-state processes and universal patterns, we combine data from the highly dynamic ecosystems of the Hawaiian Archipelago with the intentionally minimalist steady state predictions of the Maximum Entropy Theory of Ecology.

The Hawai‘i Chronosequence

The Hawaiian Archipelago holds numerous examples of adaptive radiation (Baldwin & Sanderson 1998; Bennett & O'Grady 2013; Gillespie 2004; Givnish *et al.* 2009; Knope *et al.* 2012; Lerner *et al.* 2011; Magnacca & Price 2015). Those remarkable examples of diversification along with the geographic isolation and chronological geology of the archipelago have made it a focal point for the study of eco-evolutionary processes (Barton *et al.* 2021; Chadwick *et al.* 1999; Gillespie & Clague 2009; Price & Wagner 2004; Simon 1987; Wagner & Funk 1995). The manner in which the islands formed in chronological sequence Figure 1 has allowed for the study of ecosystems in different stages of eco-evolutionary assembly (Gillespie 2004; Rominger *et al.* 2016; Vitousek 2004) by considering ecosystems on different aged substrates to be snapshots of eco-evolutionary processes at different time points.

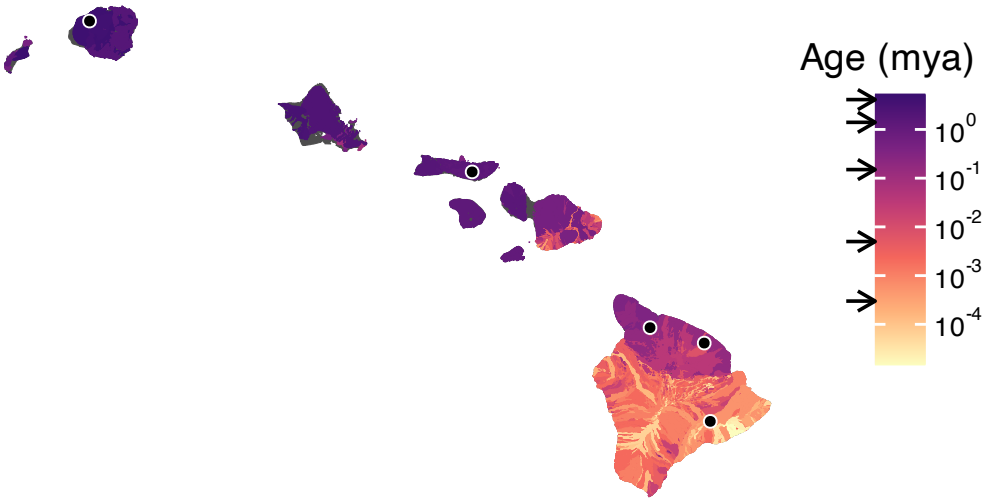


Figure 1: Chronosequence of the Hawaiian Archipelago showing oldest flows to the west on Kaua‘i and youngest flows to the east on Hawai‘i. Substrate age data from Sherrod *et al.* (2021). The five sites are indicated with dots on the map and arrows indicating their flow ages on the legend. From oldest to youngest (west to east) sites are: Alaka‘i, Kamakou, Kohala, Laupāhoehoe, Kīlauea.

Hawai‘i has also been subjected to pervasive anthropogenic changes to its biodiversity, especially a rapidly expanded human land-use footprint and markedly accelerated introduction of non-native species following Western contact and settler colonialism (Gon *et al.* 2018; Graham *et al.* 2017). As such Hawai‘i has produced numerous studies of rare and endangered species (Magnacca 2007; Pratt *et al.* 2009; Régnier *et al.* 2015; Sakai *et al.* 2002) and the functions of non-native species in ecosystems (Barton *et al.* 2021; Ostertag *et al.* 2009).

Arthropods in Hawai‘i represent both many adaptive radiations (Bennett & O'Grady 2013; Gillespie 2004; Magnacca & Price 2015), many rare species (Magnacca 2007; Magnacca & Price 2015), and many introduced species (Nishida 1992). Given their abundance and diversity, they

also present a unique opportunity to use ecological theory to study how biodiversity conforms to or deviates from steady state.

The Maximum Entropy Theory of Ecology

The Maximum Entropy Theory of Ecology (METE Harte (2011)) is a particularly useful tool for studying steady state biodiversity patterns because it produces multiple testable predictions, making falsifications of the theory more informative (McGill 2003) including the distribution of abundances across species and the distribution of metabolic rates across individuals in a sample. METE arrives at some similar predictions to other null or neutral models of biodiversity such at the Unified Neutral Theory (Hubbell 2001) but instead of being limited to a single neutral formulation, METE is grounded in the principles of statistical mechanics Jaynes (1957) drawing from the probabilistic properties of large, randomly assembled systems. Thus its predictions represent a community in statistical steady state. Statistical steady state means that a system is governed by only a few simple state variables, which constitute a state space, and that no additional processes limit the system's ability to freely explore this state space. The state variables of METE are total number of species S_0, total number of individuals N_0, and summed metabolic rate across all individuals E_0 (Harte 2011). METE posits that if state variables are in steady state, all biodiversity patterns are statistical outcomes from those state variables independently of the specific biological mechanisms governing the values of the state variables, so long as those mechanisms do not constitute information beyond the state variables themselves. Examples of such additional information could be rapid environmental change leading to mismatched niches with local environments, rapid diversification, or biological invasion. Under such scenarios the simple state variables of S_0, N_0, and E_0 would not likely constitute enough information to yield accurate predictions for biodiversity patterns such as the species abundance distribution or distribution of metabolic rates across individuals. Thus finding deviations between observed patterns of abundance and metabolic rate distributions compared to their METE predictions allows us to identify ecological systems out of statistical steady state (Harte 2011; Newman *et al.* 2020; Rominger *et al.* 2016).

In this study of arthropods communities across the Hawai‘i chronosequence, we find consistent deviations from steady state at middle aged sites. Deviations are predicted by within site β-diversity which peaks at middle aged sites. This suggests these communities are unstable, perhaps due to their age being the right time for adaptive radiation to take off for arthropods (Bennett & O'Grady 2013; Gillespie 2004). Deviations are also partially predicted by proportion of non-native species, except that the proportion of non-native species is high at the youngest site while this site also shows some of the lowest deviations from METE. This suggests the role of biological invasion is context dependent, at times producing stead state and at others driving deviation from it. More work connecting deviations from steady state to ecological and evolutionary drivers will be needed to further test these possibilities.

Methods

Reproducibility and data preparation

A custom R package, *HawaiiArthMETE* (Rominger & Thai 2026), accompanies this paper and can be used to reproduce all results. All data analyzed here are available via the package as described in Supplementary Section **??**.

Arthropod data come from the original publication by Gruner (2007), a study that collected 17024 specimens across the Hawai'i chronosequence Figure 1. Study sites in Gruner (2007) where standardized to be climatically homogeneous, seeking to capture differences in arthropod assemblages due to substrate age and not other abiotic factors.

Each specimen was identified to the species level (morpho species were used in 31.56% of specimens). Specimen lengths were measured to the nearest millimeter. For METE analyses we need to estimate metabolic rate from these data. Because METE normalizes metabolic rates such that the lowest possible rate is 1 we use a simplified 3/4 scaling rule to approximate metabolic rate B from mass M as $B \propto M^{3/4}$.

This leaves the estimation of mass from length. Gruner (2007) used taxon- and stage- specific length-to-biomass allometric scaling relationships (Gruner 2003) to approximate mass as the expected value of a log-log regression

$$\begin{aligned} \text{mass} &= e^{\widehat{\log(\text{mass})}} \\ \widehat{\log(\text{mass})} &= \log(a) + b \cdot \log(\text{length}) \end{aligned}$$

where a and b are estimated from data collected by Gruner (2003) and vary by taxon and stage.

Using these expected value masses based on length measured to the nearest milliliter yields metabolic rate approximations that are artificially discretization (see the appendix "Sampling biomass estimates from allometric equations"). These descretizations in turn lead to spurious conclusions about the predicitiveness of METE because METE assumes metabolic rates are continuous (see Supplementary Section **??**).

Therefore, instead of estimating mass as the expected value from an allometric relationship we sample masses as random draws from the full model relating mass to length

$$\begin{aligned} \text{mass} &= e^{y} \\ y &\sim \mathcal{N}(\widehat{\log(\text{mass})},\ \sigma) \\ \widehat{\log(\text{mass})} &= \log(a) + b \cdot \log(\text{length}) \end{aligned}$$

where σ is the taxon- and stage-specific standard deviation of residuals. The full procedure for sampling masses is detailed in the appendix "Sampling biomass estimates from allometric equations" and the consequences of using random draws versus expected values on the METE analysis is thoroughly investigated in Supplementary Section **??**. We conclude that using random draws for mass estimates is largely similar to using expected values, and our overall conclusions are unchanged, but using draws avoids both spurious deviations from and conformations to METE that would be introduced by using expected values of mass.

METE analysis

To produce METE predictions for the SAD and IPD we use the *meteR* R package (Rominger & Merow 2017). We compare these predictions to observed SAD and IPD patterns using a z^2-value. The use of z^2-values is discussed in Rominger & Merow (2017), but briefly: the z^2-value compares the log likelihood of the data given them METE model to the log likelihood of a hypothetical data set drawn from the METE model itself. This comparison is done many times and the actual z^2-value is calculated as

$$z^2 = \left(\frac{\log(\mathcal{L}(\text{data} \mid \text{METE})) - \hat{\mu}_{\log \mathcal{L}_{\text{METE}}}}{\hat{\sigma}_{\log \mathcal{L}_{\text{METE}}}} \right)^2$$

where $\hat{\mu}_{\log \mathcal{L}_{\text{METE}}}$ is the mean of the log likelihood of data drawn from METE itself, and $\hat{\sigma}_{\log \mathcal{L}_{\text{METE}}}$ is the standard deviation of the log likelihood of data drawn from METE itself.

If METE is truly correct in describing the data, z^2 will asymptotically have a χ^2 distribution with $d.f. = 1$ (Rominger & Merow 2017). If METE does not truly describe the data, the z^2-value is likely to be large.

We also quantify the difference between METE predictions and observed SADs and IPDs by comparing the tails of the observed and predicted distributions. These tails can be especially informative about the nature of deviations (Harte 2011). For the SAD we compare rarity as the log ratio of observed proportion of singletons to the predicted proportion of singletons, and we compare dominance as log ration of observed total abundance of the top three most abundant species to the predicted total abundance of the top three most abundant species. For the IPD we compare metrics similar to rarity and dominance: the log ratio of observed proportion of individuals with a scaled metabolism of less than 3 to its prediction, and the log ratio of the observed sum of metabolic rates for the top 10 individuals to its prediction.

Statistical analysis of deviations from METE and potential explanatory variables

We first evaluate how deviations from METE, as measured by z^2-values, vary across the chronosequence. We then test for relationships between z^2-values and proportion of non-native species at a site as well as within site β-diversity.

For all these analyses we fit linear models predicting log-transformed z^2-values responding to each explanatory variable: substrate age, proportion non-native species, and within site β-diversity. Log-transforming z^2-values brings them closer to the normal assumptions of a linear model. The need for log-transformation is not surprising: while z^2-values will be χ^2 distributed if METE is true, deviations from METE add more probability to larger values, and taken together will lead to right-skewed distribution of z^2-values. Log-transformation brings right-skewed random variables closer to normal.

When modeling deviations from METE, we have two statistical challenges to address: samples within sites could represent pseudo replicates necessitating a random-effects model; however, because explanatory variables are site-level characteristics, site-level random effects would be completely collinear with site-level explanatory variables.

To address these challenges we model the site-level mean and variance rather than each individual sample within each site. This is the most conservative approach but runs the risk of being under-powered because while we have 48 samples, we only have 5 sites. Modeling mean and variance is the same technique used in meta-analyses. We therefore make use of the *metafor* package (Viechtbauer 2010) in fitting testing these models.

Calculating β-diversity

To quantify within site β-diversity as the mean dissimilarity between samples within a site. We first calculate mean Bray-Curtis dissimilarity between samples and then, to standardize β-diversity measures across sites, we normalize mean dissimilarity to a null model. We use the `quasiswap_count` algorithm implemented in *vegan* (Oksanen *et al.* 2025) to generate null samples. This algorithm preserves row and column sums of a sample by species matrix and also seeks to maintain null matrix fill close to the observed fill thereby mimicking patchiness in the real data. Within each site we use 500 null sample-by-species matrices to generate null models for each site. The final measure of within site β-diversity is then calculated as a z-score

$$\text{Normalized } \beta = \frac{\text{BC}_{\text{obs}} - \mu_{\text{null}}}{\sigma_{\text{null}}}$$

where BC_{obs} is the observed mean Bray-Curtis dissimilarity between samples within a site, μ_{null} is the mean of the null distribution, and σ_{null} is the standard deviation of the null distribution.

We are also interested in understanding how native and non-native species contribute to the overall mean dissimilarity at a site. We can do this by partitioning Bray Curtis dissimilarity between native and non-native groups of species. The end result of partitioning Bray Curtis is

two dissimilarity values (one for native species, one for non-natives) that add to the total dissimilarity at a site. Supplementary Section **??** provides the mathematical and computational details of how we do this.

We also compare this Bray Curtis dissimilarity partitioning to a null model. We again use the `quasiswap_count` algorithm but must constrain the algorithm to permute only inside submatrices (one for native species, another for non-natives). This additional constraint is to ensure that the distributions of native and non-native abundances across sites are preserved in permuted and observed site-by-species matrices. We wrote custom code to do this in the *HawaiiArthMETE* package and described in Supplementary Section **??**.

Comparing observed Bray Curtis dissimilarity partitions to their permutational null counterparts yields a z-score for the partitioned dissimilarities. While partitions of Bray Curtis dissimilarity perfectly add to equal the overall dissimilarity, these z-scores will not perfectly add to the previously described overall dissimilarity z-scores because the partitioned z-scores are calculated from a null model with additional constraints (considering native and non-native species) compared to the overall z-scores.

Results

Deviations from METE across the chronosequence

Deviations from METE in the SAD an IPD are largely in agreement: samples with high z^2-values for the SAD tend to have high z^2-values for the IPD (see Supplementary Figure **??**). The z^2-values for both distributions peak at middle aged sites along the chronosequence Figure 2. For the SAD this is supported by a likelihood ratio test comparing a quadratic model of z^2-values to an intercept-only model with LRT statistic $\Lambda = 28.73$, $d.f. = 2$, and $P < 0.0001$.

The overall spread of IPD z^2-values is greater than that of SAD z^2-values. Not surprisingly, for quadratic model of IPD z^2-values across the chronosequence, the LRT statistic is $\Lambda = 5.513$ with $d.f. = 2$ and $P = 0.0635$. While the P-value for the test of the quadratic model for IPD z^2-values just misses the ≤ 0.05 significance cut off by the likelihood ratio test, the less conservative Wald test does show the quadratic term to be significant with a 95% confidence interval of [-0.154, -0.014], which does not overlap 0.

For the SAD, the primary driver of differences between METE predictions and observed patterns is that observed species abundances have both a higher proportion of singletons and a larger concentration of abundance in the most abundant species Figure 3 compared to METE. Taken together, this means that observed SADs are more uneven than those predicted by METE.

For the IPD there is not a consistent relationship between deviations and the tails of the metabolic rate distribution Figure 3. In terms of proportion of individuals with very low metabolisms, 19 samples showed theory over-predicting this proportion while 27 samples

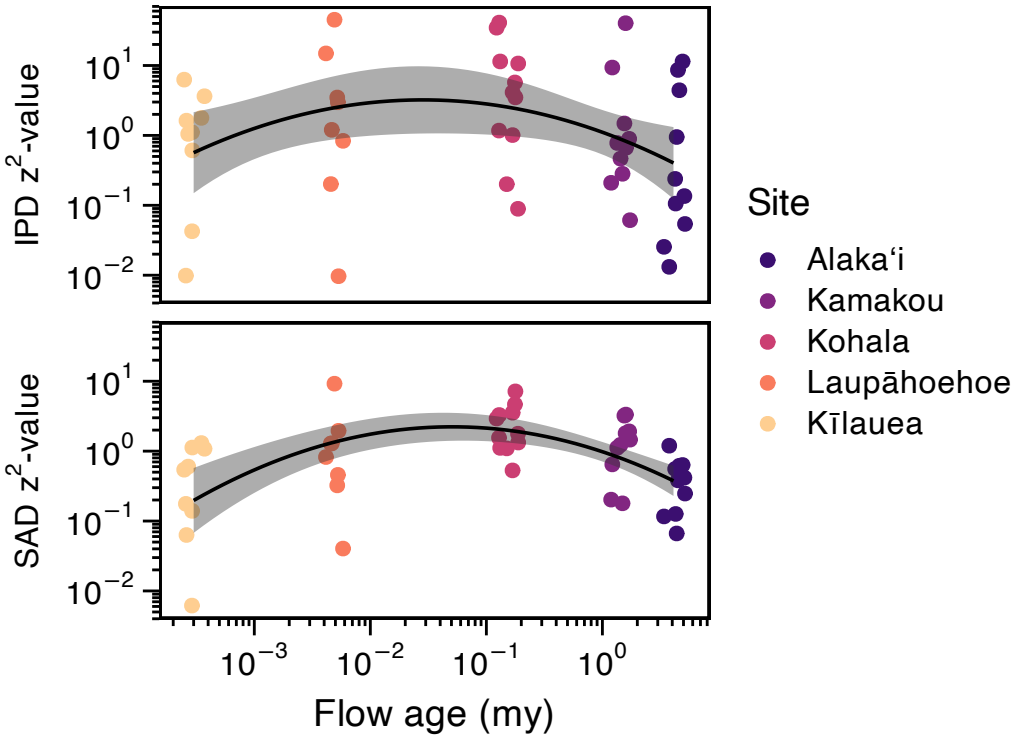


Figure 2: Deviations from METE across the chronosequence as measured by z^2-values. Colors represent sites and match the color coding of substrate age in Figure 1.

showed theory under-predicting this proportion. The difference was not significant by a binomial generalized linear model of over-versus-under prediction grouped by site (z = -1.173, P = `round(summary(p1_mod)$coefficients[4], 3)`). In terms of concentration of metabolic rate in the most metabolically active individuals, 24 samples showed theory over-predicting this concentration while 24 samples showed theory under-predicting this concentration. With the two numbers exactly equal, the difference could not be significant.

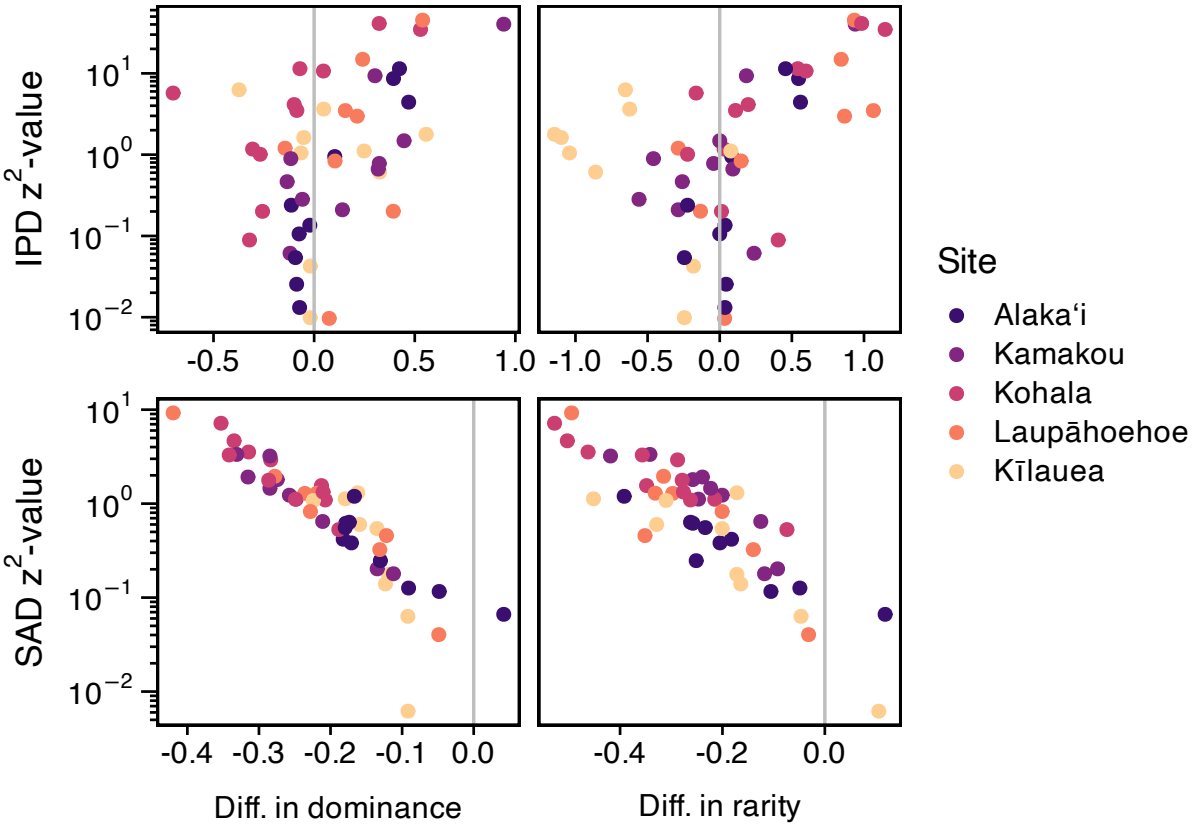


Figure 3: Relationship between z^2 measure of deviation from METE and different metrics describing how observed data and METE predictions diverge.

Deviations from METE predicted by non-native species and β-diversity

The proportion of non-native species at a site has a significant linear relationship with both SAD and IPD z^2-values when excluding Kīlauea Figure 4. For the SAD the LRT of the linear model with positive slope versus the intercept only model is $\Lambda = 22.329$ with $d.f. = 1$ and $P = < 0.001$. For the IPD the LRT of the linear model with positive slope versus the intercept only model is $\Lambda = 5.262$ with $d.f. = 1$ and $P = 0.022$.

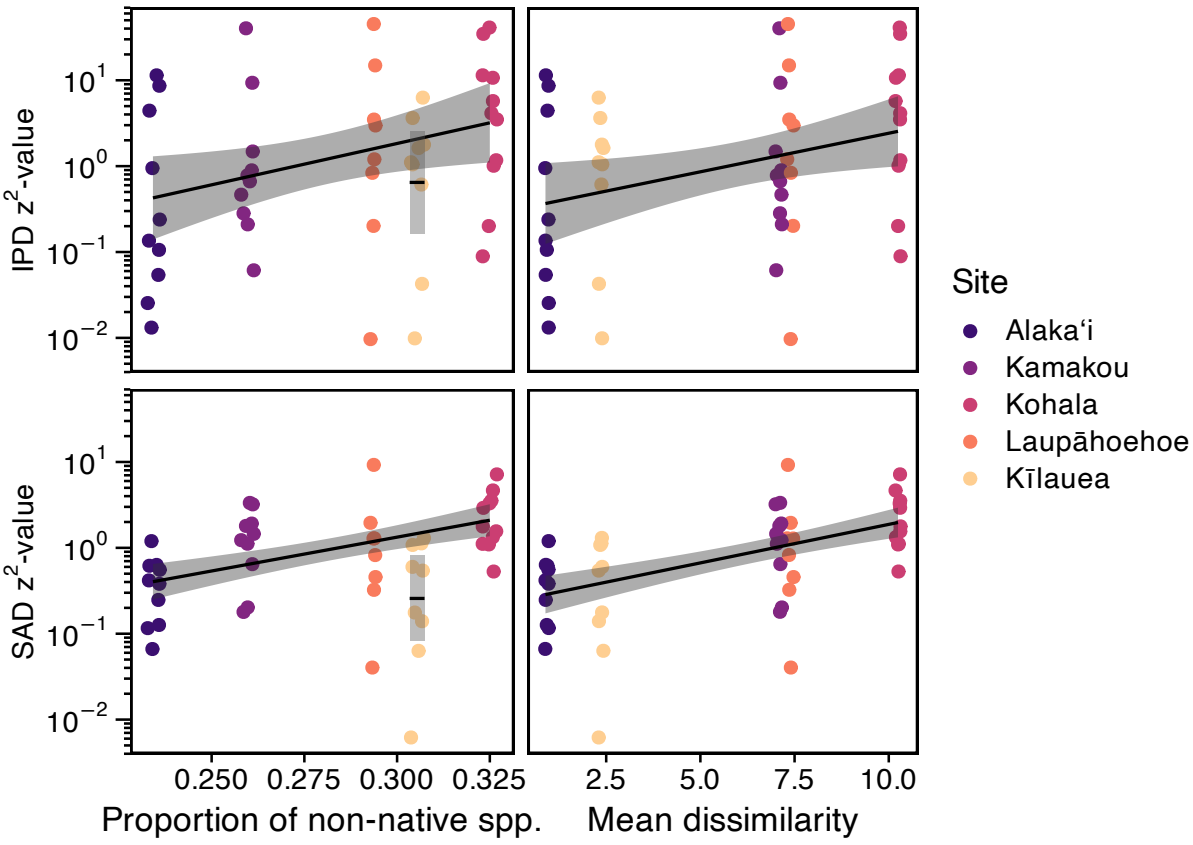


Figure 4: Relationship between z^2 measure of deviation from METE and different predictor variables: proportion of non-native species and β-diversity.

Kīlauea is an interesting outlier in both the SAD and IPD z^2-values relationships with proportion non-native species. In the case of the SAD, the 95% confidence interval for the mean of Kīlauea z^2-values is well below the 95% confidence envelope of the overall relationship Figure 4. In the case of the IPD, the 95% the mean of Kīlauea z^2-values is below the 95% confidence envelope of the overall relationship, but the confidence interval and envelope do partially overlap Figure 4. This is a strong suggestion that the effect of non-native species at Kīlauea is distinct from other sites. We unpack that possibility further in our analysis of β-diversity.

Within site β-diversity has a significant positive linear relationship with the z^2-values of both the SAD and IPD. For the SAD the LRT of the linear model with positive slope versus the intercept only model is $\Lambda = 29.968$, $d.f. = 1$, and $P < 0.0001$. For the IPD the LRT of the linear model with positive slope versus the intercept only model is $\Lambda = 5.471$, $d.f. = 1$, and $P = 0.0193$. There are no clear outlier sites in the relationship of z^2-values and β-diversity.

Partitioning β-diversity across native and non-native species

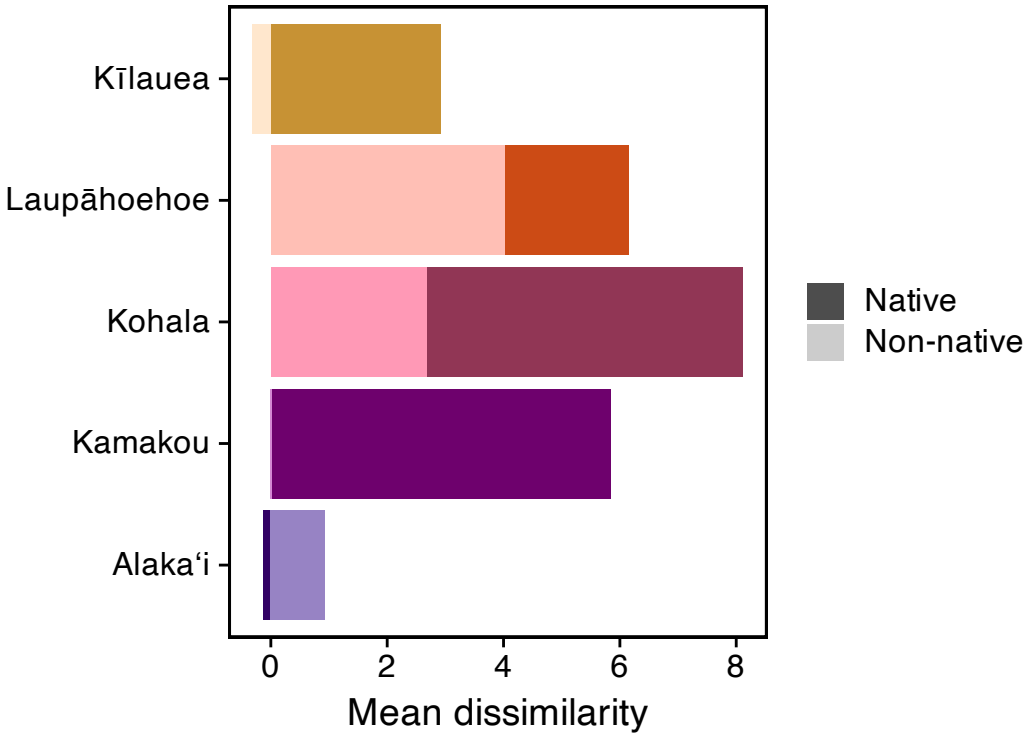


Figure 5: Difference in contribution of native and non-native species to dissimilarity compared to null. Negative values indicate greater contribution of non-native species compared to chance while positive values indicate greater contribution of native species compared to chance.

Kīlauea is an outlier in the relationship between proportion of non-native species and z^2-values but not an outlier in the relationship between β-diversity and z^2-values. To help understand why we partition native and non-native species' contributions to β-diversity. We find that while Kīlauea has a high proportion of non-native species, it also has a low β-diversity, and non-native species contribute negatively, compared to the null, to that low β-diversity Figure 5. Taking these insights together, Kīlauea has many non-native species, but those species are distributed very homogenously across samples. The oldest site Alakaʻi reverses this pattern: native species contribute negatively to the overall (low) β-diversity at that site.

The oldest (Alakaʻi) and youngest (Kīlauea) sites contrast with Laupāhoehoe, Kohala, and Kamakou sites where both native and non-native species make positive contributions to β-diversity. In Kohala and Kamakou the positive contribution of native species is substantially larger than non-natives, whereas in Laupāhoehoe the contribution of non-natives is larger.

Discussion

Real data deviate from both the METE-predicted SAD and IPD in consistent ways across the Hawaiʻi chronosequence Figure 2. This is strong (sensu McGill 2003) evidence of meaningful departures from the steady state assumptions of METE due to biological processes not accounted for in the deliberately simple state variables of the theory. Deviations in the SAD are consistently in the direction of theory under-predicting unevenness Figure 3. In one way, this result is surprising given that METE predicts a log-series distribution (Harte 2011) which is already one of the most uneven theoretical SAD shapes. However, recent extensive investigation of empirical SADs has found that deviations from statistical baselines akin to METE are often in the direction of more unevenness (Diaz *et al.* 2021). As pointed out in Diaz *et al.* (2021), deviations toward more unevenness could be accounted for by processes that increase the prevalence of rare species, and/or that increase the abundance of common species. Both niche-based (Chesson 2000; Yenni *et al.* 2012) and stochastic dispersal-based (Gotelli 1991) mechanisms could be valid causes of such processes.

Both the proportion of non-native species and the geologic age of the lava flow at a site predict deviations from METE, but in subtly different yet important ways. Across four out of five sites (all but the geologically youngest) deviations from METE increase with the proportion of non-native species Figure 4. The geologically youngest site, Kīlauea, breaks this pattern with a high proportion of non-native species but a small deviation from METE. Across all five sites deviations from METE are consistently predicted by within site β-diversity Figure 4. Both β-diversity and proportion of non-native species point to instability being a driver of deviation from METE.

The fact that geologic age seems to modulate the effect of proportion of non-native species but not within site β-diversity begs the question of how age, biological invasion, and β-diversity relate to each other. β-diversity partitions unequally between native and non-native species

across the chronosequence Figure 5. Kīlauea, the youngest site, sees non-native species contributing negatively (compared to the null) to overall β-diversity, leaving native species making the only positive contribution to overall β-diversity. The contributions of native and non-native species to the β-diversity at the oldest site Alakaʻi are reversed. But at both sites overall β-diversity is low. Thus low β-diversity seems responsible for the low deviation from METE at these sites. The high proportion of non-native species at Kīlauea is not consequential because those non-native species are very homogeneously distributed.

The opposite β-diversity contributions of native and non-native species at opposite ends of the chronosequence hints at a mechanistic possibility to explain low β-diversity and low deviation from METE. The assembly of the young Kīlauea site could be dominated by dispersal, which would be consistent with its age, and if dispersal is high, also consistent with low β-diversity accounted for by native species. Conversely, the oldest Alakaʻi site could represent the outcome of > 4 million years of eco-evolutionary assembly resulting in assemblages that are stable due to niche partitioning and maximized fitness within niches (Barabás *et al.* 2018; Chesson 2000; Godoy *et al.* 2014). This would again result in low β-diversity, with newly arriving non-native species mostly contributing to that low β-diversity, and result in steady state (i.e. low deviation from METE).

In between these mechanistic end points, our findings are consistent with the hypothesis that dispersal-assembled communities in an unstable steady state undergo a non-steady state transition via evolutionary diversification to a stable steady state. This hypothesis is consistent with our findings across the chronosequence from young to old sites. Young sites, by definition, have not had sufficient time for diversification and must be assembled by dispersal from source populations. This hypothesis is, however, only suggested by our work and would require further data collection and testing to support. In particular, extensive population genetic study would be needed to confirm the purported history of dispersal at young sites, population bifurcation leading to diversification at middle aged sites, and population stability at older sites. New molecular techniques may soon make such study a reality (Manel *et al.* 2025; e.g. Weitemier *et al.* 2021).

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